

Total No. of Questions : 6]

SEAT No. :

**P4886**

[Total No. of Pages : 2

**T.E./Insem.- 102**

**T.E. (Civil)**

**INFRASTRUCTURE ENGINEERING  
(2012 Pattern) (Semester - I)**

*Time : 1 Hour]*

*[Max. Marks : 30*

**Instructions to the candidates:**

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume suitable data if necessary.*

**Q1) a)** Discuss in detail role of infrastructure engineering in nation development. **[5]**

b) Explain the provision made for the infrastructure of airports & ports with reference to 12<sup>th</sup> five year plan. **[5]**

OR

**Q2) a)** Explain with suitable example the need of mechanization in construction sector. **[5]**

b) Discuss with suitable example the necessity of PPP projects in india. **[5]**

OR

**Q3) a)** Write short note on types of railway sleepers. **[6]**

b) Write short note on Tilting of rails. **[4]**

OR

**Q4) a)** What is ballast? Enlist various types of material used for Ballast. **[4]**

b) Define points & crossing. Draw a neat labeled diagram of turnout. **[6]**

**P.T.O.**

- Q5) a)** Write short note on slip form technique. **[4]**
- b)** What are the various methods of dewatering system? Explain any method with suitable sketch. **[6]**

OR

- Q6) a)** Define the term dredging. Explain any one type of dredging technique. **[5]**
- b)** What is hoisting? Enlist various types of hoisting equipment. **[5]**

